

* Practical - 2 *

* Write a program to implement DFS 8 Puzzle problem

* dfs.py

```
def print_in_format (matrix):
    for i in range(9):
        if i % 3 == 0 and i > 0:
            print("\n")
        print(str(matrix[i]) + " ", end = " ")
```

```
def count(s):
    ideal = [1, 2, 3,
             4, 5, 6,
             7, 8, 0]
```

```
    for i in range(9):
        if s[i] != 0 and s[i] != ideal[i]:
            c += 1
```

```
    return c
```

```
def moveCar(p, st):
    nh = 9999
    store_st = st.copy()
```

```
    for i in range(len(car)):
        dupl_st = st.copy()
```

```
        dupl_st = st.copy()
```

```
tmp = dupl_st[p]  
dupl_st[p] = dupl_st[arr[i]]  
dupl_st[arr[i]] = tmp
```

```
trh = count(dupl_st)
```

```
if trh < rh:  
    rh = trh  
    store_st = dupl_st.copy()
```

```
return store_st, rh
```

```
state = [1, 2, 3  
         0, 5, 6,  
         4, 7, 8]
```

```
h = count(state)  
level = 1
```

```
print("\n----- Level " + str(level) + "-----")  
print_in_format(state)  
print("\nHeuristic Value (Misplaced) : " + str(h))
```

```
while h > 0:  
    pos = int(state.index(0))
```

```
    level += 1
```

```
    if pos == 0:  
        arr = [1, 3]  
        state, h = move(arr, pos, state)
```



```
elif pos == 1:  
    arr = [0, 2, 4]  
    state, h = move Carrr, pos, state)
```

```
elif pos == 2:  
    arr = [1, 5]  
    state, h = move Carrr, pos, state)
```

```
elif pos == 3:  
    arr = [0, 4, 6]  
    state, h = move Carrr, pos, state)
```

```
elif pos == 4:  
    arr = [1, 3, 5, 7]  
    state, h = move Carrr, pos, state)
```

```
elif pos == 5:  
    arr = [2, 4, 8]  
    state, h = move Carrr, pos, state)
```

```
elif pos == 6:  
    arr = [3, 7]  
    state, h = move Carrr, pos, state)
```

```
elif pos == 7:  
    arr = [4, 6, 8]  
    state, h = move Carrr, pos, state)
```

```
elif pos == 8:  
    arr = [5, 7]  
    state, h = move Carrr, pos, state)
```

```
print C "\n - - - Level " + str(Clevel) +  
      " - - - - " )  
print _In_format(Cstate )  
print C "\n Heuristic Value (Misplaced) : "  
      + str(h) )
```

* Output *

- - - - Level 1 - - - -

1	2	3
0	5	6
4	7	8

Heuristic Value (Misplaced) : 3

- - - - Level 2 - - - -

1	2	3
4	5	6
0	7	8

Heuristic Value (Misplaced) : 2

----- Level 3 -----

1	2	3
4	5	6
7	0	8

Heuristic Value (Misplaced) : 1

----- Level 4 -----

1	2	3
4	5	6
7	8	0

Heuristic Value (Misplaced) : 0